A 6^r-Scaled Family of the Reduced $\lfloor 6k/5 \rfloor + 1$ Collatz-Type Map

Exact Algebraic Conjugacy, Floor-Form Representation, 5-Adic Reduction Invariance, Cycle and Inverse-Tree Transport, and Transfer of the 10^{11} Finite Verification

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Abstract

We develop the 6^r-scaled family associated with the reduced map generated by $G_0(n) = \lfloor 6n/5 \rfloor + 1$ and complete removal of factors of 5. Writing $c(n) = 5 - (n \bmod 5)$ with the convention $c(n) = 5$ when $5 \mid n$, one has $G_0(n) = (6n + c(n))/5$. For every integer $r \geq 0$ we introduce the scaled lattice $D_r = 6^r D$, where $D = \{n \geq 1 : 5 \nmid n,\ 6 \nmid n\}$ is the closed post-transient domain of the base system, and define

$$H_r(x) = \frac{6x + 6^r c(x)}{5}, \qquad T_r(x) = \frac{H_r(x)}{5^{\nu_5(H_r(x))}}.$$

Because $6^r \equiv 1 \pmod 5$ and $5 \nmid 6^r$, multiplication by 6^r preserves the residue branch and contributes no 5-adic valuation. We prove the exact conjugacy

$$T_r(6^r n) = 6^r T_0(n), \qquad T_r^j(6^r n) = 6^r T_0^j(n).$$

The generalized initial quotient also has the requested floor representation

$$H_r(x) = \left\lfloor \frac{6x}{5} \right\rfloor + 1 + \frac{6^r - 1}{5}\Big[5\big(\lfloor 6x/5 \rfloor + 1\big) - 6x\Big]$$

for $5 \nmid x$, with the bracket selecting the appropriate coefficient in $\{1, 2, 3, 4\}$. Consequently the complete directed functional graph on the scaled domain, the period-four attractor, cycle periods, predecessor trees, valuation words, first-descent times, elementary-step counts, peak positions, and normalized peak ratios are transported exactly. Absolute states and peaks are multiplied by 6^r. The completed base first-descent verification through 10^{11} therefore transfers, without a new 10^{11}-start scan, to the corresponding scaled set at every fixed r. This is an exact theorem on corresponding scaled states, not an independent exhaustive computation at each scale. The global convergence question remains equivalent to the unresolved base problem.

1 Introduction

The base reduced floor map is defined by

$$G_0(n) = \left\lfloor \frac{6n}{5} \right\rfloor + 1, \qquad T_0(n) = \frac{G_0(n)}{5^{\nu_5(G_0(n))}}.$$

A completed finite computation for the base system exhaustively treated every start $1 \leq n \leq 10^{11}$ and established strict first descent for every $n \geq 2$ in that interval, hence entry into the exact four-cycle

$(1, 2, 3, 4)$ by strong induction. The purpose of the present paper is structural: to determine the precise dynamical meaning of replacing the residue correction $1, 2, 3, 4$ by $6^r, 2 \cdot 6^r, 3 \cdot 6^r, 4 \cdot 6^r$.

The key point is that the generalized affine expression becomes a true scaled copy of the base dynamics only on the corresponding scaled lattice. On that domain, the factor 6^r can be pulled through the numerator and survives complete 5-adic reduction. This yields an exact conjugacy rather than a heuristic similarity.

We distinguish throughout: (i) exact algebraic statements valid for all $r \geq 0$; (ii) finite results inherited from the completed base computation; and (iii) probabilistic interpretations inherited only after normalization. No fresh 10^{11}-start scan is claimed for each scale.

2 The base reduced floor map

Write $n = 5q + s$, $0 \leq s \leq 4$, and define

$$c(n) = \begin{cases} 5, & s = 0, \\ 5 - s, & s = 1, 2, 3, 4. \end{cases}$$

Then

$$G_0(n) = \frac{6n + c(n)}{5} = \left\lfloor \frac{6n}{5} \right\rfloor + 1. \tag{1}$$

Let $e_0(n) = \nu_5(G_0(n))$ and

$$T_0(n) = G_0(n)/5^{e_0(n)}.$$

The base study proves that after one iteration every orbit lies in

$$D = \{m \geq 1 : 5 \nmid m,\ 6 \nmid m\},$$

and $T_0(D) \subseteq D$. On D, $c(n) \in \{1, 2, 3, 4\}$. The exact cycle

$$C = (1, 2, 3, 4), \qquad 1 \mapsto 2 \mapsto 3 \mapsto 4 \mapsto 1,$$

is contained in D.

The completed base certificate covers 10^{11} starting values. Its headline records used later are shown in Table **??**.

Table 1: Base finite-verification record inherited from the completed base computation.

Quantity	Base result
Starts tested	$100,000,000,000$
Maximum first-descent time	422
First start attaining it	$55,335,461,598$
Maximum elementary count	470
Maximum observed extra valuation	16
Largest pre-descent state	$8,669,800,813,122,806,007,604$
Start producing largest peak	$37,266,730,278$
Mean first-descent time	6.18073627485

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3 Definition of the 6^r-scaled family

Fix $r \geq 0$ and set

$$D_r = 6^r D = \{6^r n : n \in D\}.$$

For $x \in D_r$ define

$$H_r(x) = \frac{6x + 6^r c(x)}{5}, \tag{2}$$

and

$$e_r(x) = \nu_5(H_r(x)), \qquad T_r(x) = \frac{H_r(x)}{5^{e_r(x)}}. \tag{3}$$

Since $x \in D_r$ implies $5 \nmid x$, the correction $c(x)$ is one of $1, 2, 3, 4$. Thus (**??**) is precisely one of

$$\frac{6x + 6^r}{5}, \quad \frac{6x + 2 \cdot 6^r}{5}, \quad \frac{6x + 3 \cdot 6^r}{5}, \quad \frac{6x + 4 \cdot 6^r}{5},$$

with the coefficient forced by $x \bmod 5$.

4 Exact floor-function representation

The residue selector can itself be written using the base floor expression.

Lemma 4.1 (Floor residue selector)**.** *For $5 \nmid x$,*

$$c(x) = 5\left(\left\lfloor \frac{6x}{5} \right\rfloor + 1\right) - 6x.$$

Proof. Write $x = 5q + s$ with $s \in \{1, 2, 3, 4\}$. Then $\lfloor 6x/5 \rfloor = 6q + s$, so the right side is $5(6q + s + 1) - 6(5q + s) = 5 - s = c(x)$. □

Theorem 4.2 (Generalized floor form)**.** *For $x \in D_r$,*

$$H_r(x) = \left\lfloor \frac{6x}{5} \right\rfloor + 1 + \frac{6^r - 1}{5}\, c(x) \tag{4}$$

$$= \left\lfloor \frac{6x}{5} \right\rfloor + 1 + \frac{6^r - 1}{5}\left[5\left(\left\lfloor \frac{6x}{5} \right\rfloor + 1\right) - 6x\right]. \tag{5}$$

Proof. From (**??**), $(6x + c(x))/5 = \lfloor 6x/5 \rfloor + 1$. Therefore

$$\frac{6x + 6^r c(x)}{5} = \frac{6x + c(x)}{5} + \frac{(6^r - 1)c(x)}{5}.$$

Since $6^r \equiv 1 \pmod 5$, the second coefficient is integral. Lemma **??** gives (**??**). □

At $r = 0$ the correction term vanishes and the base floor map is recovered exactly.

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5 Scaling lemmas

Lemma 5.1 (Residue preservation)**.** *For every $n \in D$ and $r \geq 0$,*

$$6^r n \equiv n \pmod 5, \qquad c(6^r n) = c(n).$$

Proof. Since $6 \equiv 1 \pmod 5$, also $6^r \equiv 1 \pmod 5$. □

Lemma 5.2 (Exact affine factorization)**.** *For every $n \in D$,*

$$H_r(6^r n) = 6^r G_0(n).$$

Proof. By Lemma **??**,

$$H_r(6^r n) = \frac{6(6^r n) + 6^r c(n)}{5} = 6^r \frac{6n + c(n)}{5}.$$

□

Lemma 5.3 (5-adic valuation invariance)**.** *For every nonzero integer m,*

$$\nu_5(6^r m) = \nu_5(m).$$

Consequently $e_r(6^r n) = e_0(n)$.

Proof. The scale factor 6^r is coprime to 5. □

Corollary 5.4 (Scaled closure)**.** $T_r(D_r) \subseteq D_r$.

6 Exact conjugacy and functional-graph isomorphism

Define $\Phi_r : D \to D_r$ by $\Phi_r(n) = 6^r n$.

Theorem 6.1 (One-step conjugacy)**.** *For every $r \geq 0$ and $n \in D$,*

$$T_r(6^r n) = 6^r T_0(n), \qquad T_r \circ \Phi_r = \Phi_r \circ T_0.$$

Proof. Combine Lemmas **??** and **??**. □

Theorem 6.2 (All iterates)**.** *For every $j \geq 0$,*

$$T_r^j(6^r n) = 6^r T_0^j(n).$$

Proof. Induction on j using Theorem **??**. □

Corollary 6.3 (Mutual conjugacy of scale levels)**.** *For $r, s \geq 0$, the map $\Psi_{r,s}(6^r n) = 6^s n$ is a bijection $D_r \to D_s$ satisfying $T_s \circ \Psi_{r,s} = \Psi_{r,s} \circ T_r$.*

Thus the directed functional graphs are isomorphic after normalization.

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7 The scaled four-cycle and cycle transport

Corollary 7.1 (Scaled attractor cycle)**.** *The base cycle transports to*

$$C_r = (6^r, 2 \cdot 6^r, 3 \cdot 6^r, 4 \cdot 6^r),$$

with the same minimal period four.

Every base periodic orbit corresponds bijectively to a scaled periodic orbit of the same minimal period, and conversely.

Suppose $x_0, \dots, x_{L-1}$ is a period-L orbit in D_r. Put $a_i = 1 + e_r(x_i)$ and $c_i = c(x_i)$. Then

$$5^{a_i} x_{i+1} = 6x_i + 6^r c_i.$$

Let $A_j = \sum_{i=0}^{j-1} a_i$, $A_0 = 0$, and $A = A_L$.

Theorem 7.2 (Scaled cycle identity)**.** *Every period-L orbit satisfies*

$$(5^A - 6^L)x_0 = 6^r \sum_{j=0}^{L-1} 6^{L-1-j} c_j 5^{A_j}.$$

Writing $x_0 = 6^r n_0$ reduces this identity exactly to the base cycle identity.

Proof. Successive substitution gives the identity; division by 6^r gives the normalized form. □

Since the right side is positive, every periodic orbit still requires $5^A > 6^L$.

8 Inverse branches and predecessor-tree transport

Let $y \in D_r$ and suppose $T_r(x) = y$ with full denominator exponent $a \geq 1$. Then

$$5^a y = 6x + 6^r c, \qquad x = \frac{5^a y - 6^r c}{6}, \quad c \in \{1, 2, 3, 4\}.$$

Writing $y = 6^r z$ yields

$$x = 6^{r-1}(5^a z - c).$$

Membership $x \in D_r$ is equivalent to divisibility of $5^a z - c$ by 6, exactly the normalized base inverse-branch condition. Hence multiplication by 6^r gives a bijection between the complete predecessor tree of z and that of $6^r z$ inside the scaled domains.

9 Orbit statistics and normalized geometry

Let $n_j = T_0^j(n)$ and $x_j = T_r^j(6^r n) = 6^r n_j$. Every order relation is preserved. Therefore first-descent time is invariant, first-lower values scale by 6^r, elementary-step counts are invariant because the 5-adic valuation word is unchanged, and peak positions are invariant. Moreover

$$\max_{j \leq J} x_j = 6^r \max_{j \leq J} n_j, \qquad \frac{x_j}{x_0} = \frac{n_j}{n_0}.$$

Thus absolute peaks scale by 6^r, while peak/start ratios are identical.

Theorem 9.1 (Homogeneous statistic transport)**.** *If a finite-trajectory statistic F satisfies $F(cz_0, \dots, cz_J) = c^d F(z_0, \dots, z_J)$, then corresponding scaled and base segments satisfy*

$$F(x_0, \dots, x_J) = 6^{rd} F(n_0, \dots, n_J).$$

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10 5-adic density and drift preservation

The base map has the exact natural residue-density law

$$\Pr(e = k) = \frac{4}{5^{k+1}}, \qquad k \geq 0,$$

with $\mathbb{E}[e] = 1/4$. On the sparse lattice D_r, the appropriate comparison is relative density after normalization $x = 6^r n$. Since the valuation is preserved pointwise, every fixed finite valuation condition has exactly the same relative density at every scale.

For a matched step,

$$\frac{T_r(x)}{x} = \frac{6^r T_0(n)}{6^r n} = \frac{T_0(n)}{n}.$$

Hence normalized one-step multipliers and logarithmic increments are identical, not merely asymptotically similar. The base residue model therefore retains the same approximate multiplier $6/5^{e+1}$, arithmetic mean 1, and negative mean logarithmic drift $\log 6 - (5/4)\log 5$ at every scale after normalization.

11 Transfer of the exhaustive verification through 10^{11}

For fixed r define the corresponding scaled finite set

$$S_r(10^{11}) = \{6^r n : 1 \leq n \leq 10^{11}\}.$$

The base computation established that every base start in this interval enters C and every $n \geq 2$ has a strict first descent in at most 422 macro steps.

Theorem 11.1 (Finite verification transfer)**.** *For every fixed $r \geq 0$, every state $x = 6^r n$ with $1 \leq n \leq 10^{11}$ follows the base trajectory multiplied by 6^r. In particular it enters C_r, and for $n \geq 2$ its first-descent time is at most 422 macro steps.*

Proof. Apply all-iterate conjugacy to each already verified base start. Strict inequalities are preserved by multiplication by the positive constant 6^r. □

Remark 11.2. Theorem **??** is not a claim that 10^{11} trajectories were independently recomputed for each r. It is an algebraic transfer of a completed finite computation through a proved conjugacy.

The principal inherited records therefore become

Statistic	Invariant quantity	Scaled magnitude
First-descent record	422 steps	start $6^r(55,335,461,598)$
Elementary record	470 steps	start $6^r(55,335,461,598)$
Deepest observed valuation	$e = 16$	start $6^r(88,717,666,380)$
Largest pre-descent peak	peak position invariant	$6^r(8,669,800,813,122,806,007,604)$
Peak-producing start	—	$6^r(37,266,730,278)$

The normalized peak/start ratio is unchanged.

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12 Explicit scaled trajectories

The base cycle itself gives the simplest exact family:

$$1 \to 2 \to 3 \to 4 \to 1$$

becomes

$$6^r \to 2 \cdot 6^r \to 3 \cdot 6^r \to 4 \cdot 6^r \to 6^r.$$

For $r = 1$ this is $6 \to 12 \to 18 \to 24 \to 6$; for $r = 2$, $36 \to 72 \to 108 \to 144 \to 36$. These examples are direct consequences of the new definition and do not require independent search.

13 Why the scaled domain is essential

The expression $(6x + c(x)6^r)/5$ can be evaluated for many integers outside D_r, but the conjugacy theorem requires $x = 6^r n$. Without this condition there is no factorization

$$6x + 6^r c(x) = 6^r(6n + c(n)).$$

Therefore the generalized affine formula on arbitrary integers should not be identified with a base orbit multiplied by 6^r.

Example 13.1. Take $r = 1$ and $x = 1$, which is not in $D_1 = 6D$. Since $c(1) = 4$,

$$H_1(1) = \frac{6 + 24}{5} = 6,$$

whereas $6T_0(1) = 12$. Thus the scaled-domain hypothesis is necessary.

14 Global convergence and counterexample transport

Conjecture 14.1 (Base global convergence)**.** *Every positive integer eventually enters C under the base reduced map.*

Conjecture 14.2 (Scaled global convergence)**.** *For fixed r, every state in D_r eventually enters C_r.*

Theorem 14.3 (Equivalence across scales)**.** *Base global convergence on D holds if and only if scaled global convergence holds on D_r for any fixed r. Nontrivial cycles and nonconvergent orbits likewise transport bijectively between scale levels.*

This theorem preserves the unresolved status of the global problem: scaling neither proves nor disproves the base conjecture.

15 Counting at corresponding cutoffs

Let $A_0(X) = \#\{n < X : n \in D\}$ and $A_r(Y) = \#\{x < Y : x \in D_r\}$. The bijection gives

$$A_r(6^r X) = A_0(X), \qquad A_r(Y) = A_0(Y/6^r).$$

Thus exact transfer concerns corresponding normalized cutoffs, not equal absolute cutoffs.

10.5281/zenodo.23533046 v1 [NT] 21 Oct 2026 ORCID/ GITHUB

16 Computational consistency checks and reproducibility package

The accompanying verifier checks, over a user-selected normalized range and scale levels,

$$x \bmod 5 = n \bmod 5, \quad H_r(6^r n) = 6^r G_0(n), \quad e_r(6^r n) = e_0(n), \quad T_r(6^r n) = 6^r T_0(n),$$

and independently compares the residue-affine and floor-function forms. The package also contains machine-readable definitions, the base summary and selected base record files as provenance, and SHA-256 checksums. The verifier is a consistency aid; the conjugacy proof is algebraic and does not depend on it.

17 What is proved, what is inherited, and what remains heuristic

Exact algebra. The generalized floor representation, residue preservation, affine factorization, 5-adic valuation invariance, conjugacy, all-iterate correspondence, functional-graph isomorphism, scaled cycle identity, predecessor-tree transport, orbit-statistic transport, and equivalence of global questions are exact for all $r \geq 0$ on the scaled domains.

Inherited finite theorem. The completed base scan through 10^{11} transfers exactly to $S_r(10^{11})$ for every fixed r. Record times and valuation words are invariant; magnitude records acquire the factor 6^r.

Probabilistic interpretation. The one-step residue-density law is exact, but treating successive orbit states as independent or sufficiently mixing remains heuristic. The drift discussion is explanatory and is not a proof of universal convergence.

18 Conclusion

The family generated by

$$\frac{6x + 6^r}{5}, \quad \frac{6x + 2 \cdot 6^r}{5}, \quad \frac{6x + 3 \cdot 6^r}{5}, \quad \frac{6x + 4 \cdot 6^r}{5}$$

according to the residue of x modulo 5, followed by complete 5-adic reduction, has a simple exact interpretation on $D_r = 6^r D$. The identity $6^r \equiv 1 \pmod 5$ preserves the branch, while $\gcd(6^r, 5) = 1$ preserves the complete valuation. The result is the exact conjugacy $T_r(6^r n) = 6^r T_0(n)$.

The requested floor form is given by Theorem **??**; at $r = 0$ it collapses to $\lfloor 6x/5 \rfloor + 1$. At every scale, the normalized dynamics are therefore identical to the base dynamics. The four-cycle scales to $(6^r, 2 \cdot 6^r, 3 \cdot 6^r, 4 \cdot 6^r)$; cycles and predecessor trees correspond; stopping times and valuation words are unchanged; and absolute magnitudes scale by 6^r.

The exhaustive base verification through 10^{11} consequently yields a simultaneous finite theorem on all corresponding scaled sets without repeating the massive computation. The global convergence question, however, remains exactly the base global problem in normalized coordinates.

A Compact derivation of the scaling mechanism

The entire argument is summarized by

$$6^r \equiv 1 \pmod 5 \Rightarrow c(6^r n) = c(n),$$

10.5281/zenodo.23533046 v1 [NT] 21 Oct 2026 ORCID/ GITHUB

$$6(6^r n) + 6^r c(n) = 6^r(6n + c(n)),$$
$$5 \nmid 6^r \Rightarrow \nu_5(6^r m) = \nu_5(m),$$
$$T_r(6^r n) = 6^r T_0(n) \Rightarrow T_r^j(6^r n) = 6^r T_0^j(n).$$

Everything else in the scaled-domain theory follows from these identities and the positivity of the scale factor.

B Reproducibility commands

A lightweight verification run is

```
python3 src/verify_scaled_family.py --limit 10000 --max-r 5 \
  --out data/verification_results.json
```

The distributed JSON file records a passing run with 300,000 assertions.

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